

| STAT214 Chapter 9 Continuous Responses and Least Squares

| 1 Visualizing Predictive Relationships

- visualize linear relationship using a scatterplot
 - observation says nothing about a *causal* relationship
 - use fitted line to capture the linear relationship
-

| 2 Computing Fitted Lines

| 2.1 Least Absolute Deviation

Objective function:

$$\frac{1}{n} \sum_{i=1}^n |y_i - (b_0 + b_1 x_i)|$$

Model solving:

Find *reasonable* values of b_0 and b_1 is to plug a variety of potential b_0 and b_1 values into the absolute value loss function, and then choose the values of b_0 and b_1 that lead to the smallest loss.

| 2.2 Least Squares

Objective function:

$$\frac{1}{n} \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2$$

Model solving:

$$b_0 = \bar{y} - b_1 \bar{x}$$
$$b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Relationship Between the Least Squares Coefficient and Correlation:

$$b_1 = \text{Corr}(x, y) \frac{SD(y)}{SD(x)}$$

Remark: Least Squares and Linear Regression

The LS algorithm itself does not rely on such assumptions, so we present LS as a stand-alone algorithm that can be used outside of the context and assumptions of traditional linear regression.

| 2.3 Least Squares versus Least Absolute Deviation

- the LAD algorithm is more robust (stable) to outliers than the LS algorithm.
 - the more training data points you have, the less sensitive the LS algorithm will be to outliers
-

| 3 Quantitative Measures of Predictive Performance

| 3.1 Mean Squared Error, Mean Absolute Error, and Median Absolute Deviation

mean squared error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\text{observed response}_i - \text{predicted response}_i)^2$$

root-mean squared error (rMSE)

$$\text{rMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\text{observed response}_i - \text{predicted response}_i)^2}$$

mean absolute error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\text{observed response}_i - \text{predicted response}_i|$$

median absolute deviation (MAD)

$$\text{MAD} = \text{median}(|\text{observed response}_i - \text{predicted response}_i|)$$

⚠ Remark ▾

- MSE/rMSE is often the **default** performance measure.
- these measures are not directly comparable to one another
- we generally recommend computing multiple performance measures to obtain a well-rounded view of the relative predictive performance
- they need to be interpreted relative to the scale of the response variable (and thus do not provide a stand-alone picture of an algorithm's predictive performance)

| 3.2 Correlation and R-Squared

correlation performance measure

$$\text{Corr}(\text{obs}, \text{pred}) = \frac{\sum_{i=1}^n (\text{obs}_i - \overline{\text{obs}})(\text{pred}_i - \overline{\text{pred}})}{\sqrt{\sum_{i=1}^n (\text{obs}_i - \overline{\text{obs}})^2 \sum_{i=1}^n (\text{pred}_i - \overline{\text{pred}})^2}}$$

As a general rule of thumb, a correlation of around 0.6 is **OK**, a correlation of 0.7 is **good**, a correlation of around 0.8 is **very good**, and a correlation of around 0.9 is **excellent**, but these benchmarks will depend on the context of the problem.

R-squared (Training dataset only)

$$R^2 = \text{Corr}(\text{obs}, \text{pred})^2 = \frac{(\sum_{i=1}^n (\text{obs}_i - \overline{\text{obs}})(\text{pred}_i - \overline{\text{pred}}))^2}{\sum_{i=1}^n (\text{obs}_i - \overline{\text{obs}})^2 \sum_{i=1}^n (\text{pred}_i - \overline{\text{pred}})^2}$$

- it corresponds to the proportion of the variability in the response that is explained by the features
- the R^2 measure is explicitly defined in the context of the training data
- when evaluating predictions for the validation/test dataset (or external/future data), we will stick with the **correlation** performance measure rather than R^2

Scatterplot of Observed versus Predicted Responses

you can often get a sense of the correlation between two variables by looking at a scatterplot of one versus the other.

| 4 PCS Scrutinization of Prediction Results

| 4.1 Predictability

- evaluations based on the training data do not tell us how well the algorithms would perform on **future or external** data
- The predictive performance is evaluated using the validation dataset

| 4.2 Stability

If you are a Python user, the Yu research group has created the **veridical-flow** Python library for conducting stability analysis across different data perturbations, as well as cleaning, preprocessing, and algorithmic judgment calls in the context

of predictive algorithms

4.2.1 Stability to Data Perturbations and Choice of Algorithm

we first need to decide what constitutes an appropriate perturbation:

- adding random noise
- bootstrapping (needs exchangeability)

4.2.2 Prediction Stability Plots

prediction stability plot shows the **range** (as a line segment) of the perturbed predicted responses (x -axis) against the actual observed response (y -axis) for the validation set. The shorter the perturbed prediction line segment for a data point, the more stable its perturbed predictions are.

